# Lecture 16
# Expected Value, Bernoulli and Binomial Distributions

# Review

**1 What are independent events?**

Events where one occurring does not change the probability of the other.

**2 What is a random variable?**

The outcome of a random trial recorded as a number.

**3 What is a discrete random variable?**

One whose possible values can be listed, like the number of heads in 3 tosses.

**4 What is a probability distribution?**

A function giving each value of X a probability between 0 and 1; they add to 1.

# The mean of a random variable

expected value $\mu$

**the long-run average of X: $\mu = \Sigma\, x \cdot P(x)$**

variance $\sigma^2$

**$\sigma^2 = \Sigma\, (x - \mu)^2 \cdot P(x)$, and the standard deviation is $\sigma = \sqrt{\sigma^2}$**

three coin tosses

**$\mu = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 1.5$ heads**

# Example: grade points

A STAT 251 exam grade on the 4-point scale, for a student chosen at random:

| x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| grade | F | D | C | B | A |
| P(x) | 0.05 | 0.10 | 0.30 | 0.25 | 0.30 |

**(a) What is P(better than a C)?**

**(b) What is the mean grade, μ?**

**(a) P(X = 3) + P(X = 4) = 0.25 + 0.30 = 0.55**

**(b) μ = 0(0.05) + 1(0.10) + 2(0.30) + 3(0.25) + 4(0.30) = 2.65**

# Bernoulli trials

Bernoulli trial

**one trial with two outcomes: success (X = 1) with probability p, failure (X = 0) with probability 1 − p**

X ~ Bernoulli(p)

**its single parameter p is the probability of success**

mean and variance

**$\mu = p$ and $\sigma^2 = p(1 - p)$**

# Example: guessing one question

You guess on a multiple-choice question with 4 options, one of them correct.

**X = 1 if the guess is right, X = 0 if not;  p = 1/4 = 0.25**

**(a) What is P(X = 0)?**

**(b) What are $\mu$ and $\sigma^2$?**

**(a) $1 - 0.25 = 0.75$**

**(b) $\mu = 0.25$,  $\sigma^2 = 0.25 \times 0.75 = 0.1875$**

# The binomial setting

a fixed number of trials n

**the trial is repeated a set number of times, decided in advance**

two outcomes, the same p

**each trial is a success or a failure, with the same probability of success p every time**

independent trials

**the outcome of one trial does not change the probability of success on any other**

# The binomial distribution

$X \sim \text{Binom}(n, p)$

**$X$ = the number of successes in $n$ independent trials, each with probability of success $p$**

$P(X = k)$

**$\frac{n!}{(k!(n-k)!)} \cdot p^k \cdot (1-p)^{n-k}$: the first factor counts the ways to arrange $k$ successes**

mean and variance

**$\mu = np$ and $\sigma^2 = np(1-p)$**

# Example: counting vs. the formula

Monday: X = the number of heads in 3 tosses of a fair coin, counted straight from the sample space.

**S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT};  X ~ Binom(n = 3, p = 0.5)**

**(a) From S, what are P(X = 3) and P(X = 2)?**
**(b) Use the formula to find P(X = 3) and P(X = 2).**
**(c) What does 3! / (2! 1!) count?**

**(a) 1/8 (HHH) and 3/8 (HHT, HTH, THH)**
**(b) $3!/(3!\,0!) \cdot 0.5^3 \cdot 0.5^0 = 1/8$  $3!/(2!\,1!) \cdot 0.5^2 \cdot 0.5^1 = 3/8$: the same**
**(c) the 3 orders with 2 heads: HHT, HTH, THH**

# Example: guessing a whole quiz

You guess on all 10 questions of a 4-option multiple-choice quiz.

**X ~ Binom(n = 10, p = 0.25)**

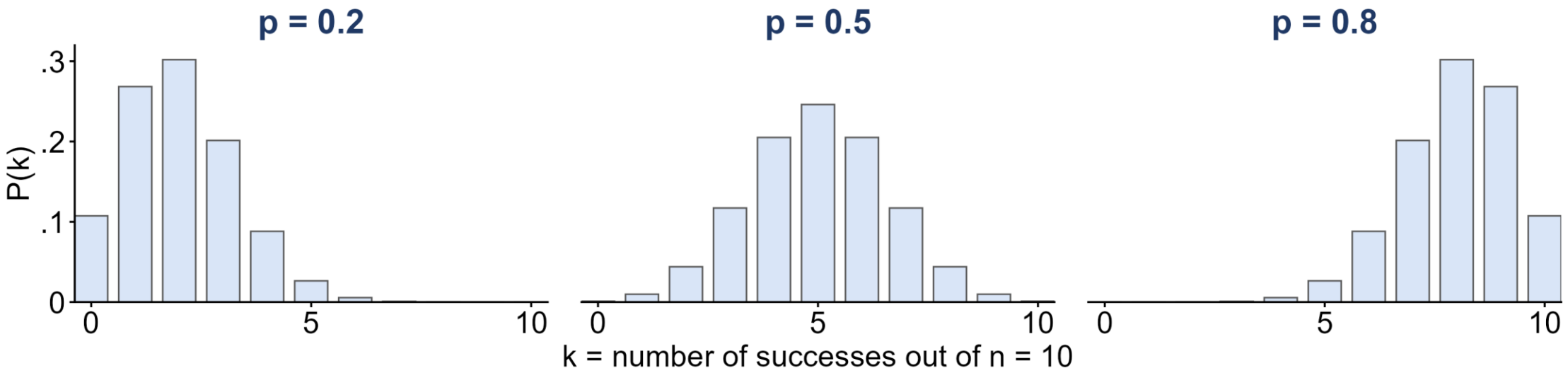
**(a) What is P(X = 3)?**
**(b) What is P(X = 0)?**
**(c) What are μ and σ?**

**(a) $10!/(3!\,7!) \cdot 0.25^3 \cdot 0.75^7 = 120 \times 0.0156 \times 0.1335 \approx 0.250$**
**(b) $0.75^{10} \approx 0.056$**
**(c) $\mu = 10 \times 0.25 = 2.5$, $\sigma = \sqrt{(10 \times 0.25 \times 0.75)} \approx 1.37$**

# The shape of a binomial distribution

X ~ Binom(n = 10, p) for three values of p.

**What does p do to the shape?**




**p = 0.5: symmetric. p < 0.5: skewed right. p > 0.5: skewed left.**
**The center sits at μ = np: 2, 5 and 8**